Algorithm 1 Update the grid hierarchy

DATA: Grid G at level l
- Update the grid G of Δt_l
- Set the boundary condition for each child grid of G
- Update each child grid $r = \frac{\Delta t_l}{\Delta t_{l+1}}$ times
- Update the values of the control volumes overlapping finer control volumes
- Correct the flux

Algorithm 2 Creation of the new grid hierarchy

DATA: Grid G at refinement level l
if The gird G does not have child grid or $l = l_{max} - 1$ **then**
- Apply the refinement criteria
- Restrict de child grids into the refinement areas
- Create new child grids and extend existing child grids to fit the refinement areas

if $l < l_{max} - 1$ **then**
Apply **Algorithm 2** to every child grid of G
end if
else
- Apply **Algorithm 2** to every child grid of G
- Apply the refinement criteria
- Create new child grids and extend existing child grids to fit the refinement areas
- Apply **Algorithm 2** to every new child grid of G

end if

Algorithm 3 Richardson extrapolation

DATA: Grid G at refinement level l, error tolerance ϵ, number of buffer zone N_b, time t

if l $= 1$ **then**

- Create the grid G_0 twice as coarse as G
- Update G_0 of Δt_0
- Update G of Δt_1 $r = \frac{\Delta t_0}{\Delta t_1}$ times
- Flag for refinement every couple of control volumes of G where $|U_I^0(t + \Delta t_0) - \frac{1}{2}(U_i^1(t + \Delta t_0) + U_{i+1}^1(t + \Delta t_0))| \geq \epsilon$

else

- Update G_{l-1} of Δt_{l-1}
- Update G of Δt_l $r = \frac{\Delta t_{l-1}}{\Delta t_l}$ times
- Flag for refinement every couple of control volumes of G where: $|U_I^{l-1}(t + \Delta t_{l-1}) - \frac{1}{2}(U_i^l(t + \Delta t_{l-1}) + U_{i+1}^l(t + \Delta t_{l-1}))| \geq \epsilon$

end if

- Flag for refinement every N_b control volumes next to an already flagged control volume

Algorithm 4 Apriori estimator

DATA: Grid G at level l
if $l < l_{max}$ **then**
  Compute $\overline{\nabla U_{G,-}}^2$ and $\overline{\nabla U_{G,-}}^2$
  if $\overline{\nabla U_{G,\pm}}^2 \geq \eta_l$ **or** a control volume of $G_\pm$ succeed the Kelvin-Helmholtz instability test **then**
    Flag for refinement the half grid $G_\pm$
  end if
  Apply **Algorithm 4** to G_+ and G_-
end if